Lecture Notes

Lecture 5

Eigenvalues and Eigenvectors

Lecture 4 ended with a tantalising example: the right basis turned a reflection into the diagonal matrix $\mathrm{diag}(1, -1)$. The diagonal entries were the factors by which the operator stretched two special directions. This lecture names those objects—*eigenvectors*, the directions an operator merely scales, and *eigenvalues*, the scaling factors—and develops the tools to find them: the characteristic polynomial, the guaranteed existence of eigenvalues over $\mathbb{C}$, and upper-triangular form, in which the eigenvalues sit on the diagonal.

The stakes for physics could hardly be higher. A quantum observable is an operator; its eigenvalues are the values a measurement can return, and its eigenvectors are the states with a definite value. The Hamiltonian's eigenvalues are the allowed energies, and the time-independent Schrödinger equation $\hat{H}\psi = E\psi$ is nothing but the statement that ψ is an eigenvector. Everything from spectral lines to the stability of matter is an eigenvalue problem.

Learning objectives

By the end of this lecture you will be able to:

- Identify invariant subspaces, eigenvalues, eigenvectors, and eigenspaces.
- Compute eigenvalues from the characteristic polynomial and recover tr and det.
- Explain why every operator over $\mathbb{C}$ has an eigenvalue (and may have none over $\mathbb{R}$).
- Show eigenvectors for distinct eigenvalues are independent; triangularise over $\mathbb{C}$.

1 Invariant subspaces and eigenvectors

Given $T \in \mathcal{L}(V)$, the cleanest way to understand T is to break V into pieces that T preserves.

Definition 5.1 (Invariant subspace). A subspace $U \subseteq V$ is *invariant* under $T \in \mathcal{L}(V)$ if $Tu \in U$ for every $u \in U$, i.e. $T(U) \subseteq U$.

The whole space V, the zero subspace $\{0\}$, the kernel $\mathrm{null}\, T$, and the range $\mathrm{range}\, T$ are always invariant. The richest information, though, comes from the smallest nontrivial invariant subspaces—the one-dimensional ones. A one-dimensional subspace $\mathrm{span}(v)$ (with $v \neq 0$) is invariant exactly when Tv is a scalar multiple of v.

Definition 5.2 (Eigenvalue, eigenvector, eigenspace). A scalar $\lambda \in \mathbb{F}$ is an *eigenvalue* of $T \in \mathcal{L}(V)$ if there is a nonzero $v \in V$ with

$$Tv = \lambda v.$$

Such a v is an *eigenvector* for λ. The set of all eigenvectors for λ together with 0 is the *eigenspace*

$$E(\lambda, T) = \mathrm{null}(T - \lambda I) = \{v \in V : Tv = \lambda v\},$$

a subspace of V. The set of all eigenvalues is the *spectrum* $\mathrm{spec}(T)$.

Because $E(\lambda, T) = \mathrm{null}(T - \lambda I)$, the following conditions on a scalar λ are equivalent, and we will use them interchangeably:

$$\lambda \in \mathrm{spec}(T) \iff T - \lambda I \text{ is not injective} \iff T - \lambda I \text{ is not invertible.} \tag{5.1}$$

(The last equivalence uses that an operator on a finite-dimensional space is injective iff invertible, from Lecture 3.) Geometrically, an eigenvector is a direction left invariant by T, merely rescaled (Figure 1).

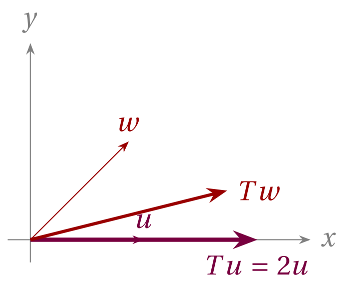

Figure 1: For the stretch $T(x, y) = (2x, \frac{1}{2}y)$, the x-axis is an eigendirection ($Tu = 2u$); a generic vector w is turned, so it is not an eigenvector.

Example 5.3 (A catalogue of eigenvalues).

- A *projection* P with $P^2 = P$ has eigenvalues among $\{0, 1\}$: $E(1, P) = \operatorname{range} P$ and $E(0, P) = \operatorname{null} P$.
- A *reflection* across a line in $\mathbb{R}^2$ has eigenvalues $+1$ (along the line) and -1 (across it), as we saw in Lecture 4.
- *Differentiation* on the space of smooth functions has every λ as an eigenvalue: $De^{\lambda t} = \lambda e^{\lambda t}$. The exponentials are its eigenvectors.
- A *rotation* R_θ of $\mathbb{R}^2$ by $\theta \notin \{0, \pi\}$ has *no* real eigenvalue—it turns every nonzero vector (Figure 2). ◁

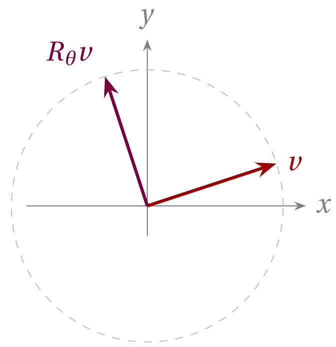

Figure 2: A rotation moves every nonzero vector off its own line, so over $\mathbb{R}$ it has no eigenvector. Over $\mathbb{C}$ its eigenvalues are $e^{\pm i\theta}$.

Example 5.4 (Computing eigenvalues of a 2×2 matrix). Let $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ act on $\mathbb{R}^2$. A scalar λ is an eigenvalue iff $A - \lambda I$ is singular, i.e. $\det(A - \lambda I) = (2 - \lambda)^2 - 1 = 0$, giving $\lambda = 1$ or $\lambda = 3$. For $\lambda = 3$, $(A - 3I)v = 0$ reads $-v_1 + v_2 = 0$, so $v = (1, 1)$; for $\lambda = 1$, $v = (1, -1)$. The two eigenvectors point along and across the line $y = x$, just as the symmetric form of A suggests. ◁

Example 5.5 (Invariant but not an eigenline). Invariance is weaker than being spanned by an eigenvector. For $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$, the x-axis $\operatorname{span}\big((1, 0)\big)$ is invariant (its vectors are eigenvectors with eigenvalue 1), and $\mathbb{R}^2$ is invariant without being one-dimensional, but the line $\operatorname{span}\big((1, 1)\big)$ is *not* invariant: $A(1, 1) = (2, 1)$, which is off that line. Only special lines are eigenlines. ◁

Example 5.6 (Complex eigenvectors: the Pauli matrix σ_y). The observable $\sigma_y = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$ is genuinely complex. Its characteristic polynomial is $\det\left(\begin{smallmatrix}\lambda & i\\ -i & \lambda\end{smallmatrix}\right) = \lambda^2 - 1$, so the eigenvalues are again ± 1, but the eigenvectors are now complex: $\frac{1}{\sqrt{2}}(1, i)$ for $+1$ and $\frac{1}{\sqrt{2}}(1, -i)$ for -1. These are the spin states pointing along $\pm y$ (the spin analogue of circular polarization); they live in $\mathbb{C}^2$ with no real representative, a reminder that the natural arena for quantum mechanics is a *complex* vector space. ◁

2 Eigenvectors for distinct eigenvalues are independent

Theorem 5.7 (Independence of eigenvectors). Let $v_1, \ldots, v_m$ be eigenvectors of T for *distinct* eigenvalues $\lambda_1, \ldots, \lambda_m$. Then $v_1, \ldots, v_m$ is linearly independent.

Proof. Suppose not, and let k be smallest such that $v_k \in \operatorname{span}(v_1, \ldots, v_{k-1})$; write $v_k = a_1 v_1 + \cdots + a_{k-1} v_{k-1}$. Apply T and separately multiply by λ_k:

$$\lambda_k v_k = \sum_{j<k} a_j \lambda_j v_j, \qquad \lambda_k v_k = \sum_{j<k} a_j \lambda_k v_j.$$

Subtracting, $0 = \sum_{j<k} a_j(\lambda_j - \lambda_k) v_j$. By minimality of k the vectors $v_1, \ldots, v_{k-1}$ are independent, so each $a_j(\lambda_j - \lambda_k) = 0$; since the eigenvalues are distinct, every $a_j = 0$, forcing $v_k = 0$—a contradiction, as eigenvectors are nonzero. ∎

Corollary 5.8 (At most $\dim V$ eigenvalues). An operator on an n-dimensional space has at most n distinct eigenvalues.

Proof. Eigenvectors for distinct eigenvalues form an independent list, which can have at most $n = \dim V$ entries. ∎

Corollary 5.9 (Eigenspaces are independent). For distinct eigenvalues $\lambda_1, \ldots, \lambda_m$ of T, the sum $E(\lambda_1, T) + \cdots + E(\lambda_m, T)$ is direct.

Proof. A nontrivial way of writing 0 as a sum $u_1 + \cdots + u_m$ with $u_j \in E(\lambda_j, T)$ and not all $u_j = 0$ would be a vanishing combination of eigenvectors for distinct eigenvalues, contradicting Theorem 5.7. Hence the only representation of 0 is trivial, so the sum is direct. ∎

If an operator on an n-dimensional space happens to have n distinct eigenvalues, the corresponding eigenvectors form a basis, and in that basis the matrix is diagonal. This is the easy route to diagonalization, taken up systematically in Lecture 6.

Example 5.10 (A diagonalizing basis). The matrix $A = \left(\begin{smallmatrix}2 & 1\\ 1 & 2\end{smallmatrix}\right)$ has the distinct eigenvalues 3, 1 with eigenvectors $(1, 1), (1, -1)$ (Example 5.4). In the basis $\mathcal{B}' = \big((1, 1), (1, -1)\big)$ the operator becomes diagonal: with $P = \left(\begin{smallmatrix}1 & 1\\ 1 & -1\end{smallmatrix}\right)$,

$$P^{-1} A P = \begin{pmatrix}3 & 0\\ 0 & 1\end{pmatrix} = \operatorname{diag}(3, 1),$$

the eigenvalues displayed on the diagonal. Distinct eigenvalues always furnish such a basis. ◁

3 Existence of eigenvalues over $\mathbb{C}$

Over $\mathbb{R}$ an operator may have no eigenvalue at all (the rotation). Over $\mathbb{C}$ the situation is completely different, and the reason is the fundamental theorem of algebra.

> **Theorem 5.11 (Eigenvalues exist over $\mathbb{C}$).** Every operator on a finite-dimensional, nonzero, complex vector space has an eigenvalue.

Proof. Let $\dim V = n \geq 1$ and pick any $v \neq 0$. The $n+1$ vectors $v, Tv, T^2v, \ldots, T^nv$ cannot be independent, so some nontrivial combination vanishes:

$$a_0 v + a_1 Tv + \cdots + a_n T^n v = 0,$$

with not all a_k zero (and the top nonzero coefficient defining a polynomial of degree $m \geq 1$). Factor the corresponding polynomial over $\mathbb{C}$ using the fundamental theorem of algebra:

$$a_0 + a_1 z + \cdots + a_m z^m = c\,(z - \lambda_1)\cdots(z - \lambda_m), \qquad c \neq 0.$$

Substituting T for z and applying to v,

$$0 = c\,(T - \lambda_1 I)\cdots(T - \lambda_m I)\,v.$$

The product of operators kills $v \neq 0$, so at least one factor $T - \lambda_j I$ is not injective. That λ_j is an eigenvalue of T. ■

Remark 5.12. This is the deepest reason quantum mechanics is built over $\mathbb{C}$: it guarantees that observables have eigenvalues—possible measurement outcomes—and, with the extra structure of Lecture 9, that there are enough eigenvectors to form a basis. Over $\mathbb{R}$ no such guarantee exists.

Example 5.13 (The existence proof in action). Take $T(z_1, z_2) = (z_2, -z_1)$ on $\mathbb{C}^2$. Starting from $v = (1, 0)$ we get $Tv = (0, -1)$ and $T^2v = (-1, 0) = -v$, so $(T^2 + I)v = 0$. The polynomial $z^2 + 1 = (z - i)(z + i)$ factors over $\mathbb{C}$, hence $(T - iI)(T + iI)v = 0$; since $v \neq 0$ one factor is singular, giving the eigenvalues $\pm i$. Over $\mathbb{R}$ the polynomial would not factor and the construction would stall—which is exactly why this real rotation has no real eigenvalue. ◁

> **Proposition 5.14 (Commuting operators share an eigenvector).** If V is a nonzero finite-dimensional complex vector space and $S, T \in \mathcal{L}(V)$ commute, then S and T have a common eigenvector.

Proof. Let λ be an eigenvalue of S (Theorem 5.11). The eigenspace $E(\lambda, S)$ is invariant under T: if $Sv = \lambda v$ then $S(Tv) = T(Sv) = \lambda(Tv)$, so $Tv \in E(\lambda, S)$. The restriction of T to the nonzero complex space $E(\lambda, S)$ has an eigenvector v (Theorem 5.11 again), and this v is an eigenvector of T by construction and of S because it lies in $E(\lambda, S)$. ■

This is the abstract statement behind *compatible observables*: commuting operators can be simultaneously diagonalized, so they admit states of simultaneously definite values, whereas non-commuting ones cannot.

4 The characteristic polynomial

Equation (5.1) says eigenvalues are the scalars making $T - \lambda I$ singular. In coordinates, singularity is detected by a determinant.

Remark 5.15 (Determinant facts we use). Beyond the cofactor expansion of your prerequisites, this course freely uses four facts about determinants, recorded here for reference: (i) a square matrix is invertible iff $\det A \neq 0$; (ii) the product rule $\det(AB) = \det A \det B$, whence $\det(P^{-1}AP) = \det A$, so the determinant is a similarity invariant and belongs to the operator; (iii) the determinant of a triangular matrix is the product of its diagonal entries; (iv) expanding $\det(\lambda I - A)$ in powers of λ gives $\lambda^n - (\operatorname{tr} A)\lambda^{n-1} + \cdots + (-1)^n \det A$. Each follows from the cofactor expansion (or row reduction) and is proved in any of the course texts; we take them as given.

Definition 5.16 (Characteristic polynomial). For $A \in M_n(\mathbb{F})$ (or for $T \in \mathcal{L}(V)$ via any basis), the *characteristic polynomial* is

$$\chi_A(\lambda) = \det(\lambda I - A).$$

It is monic of degree n, and its roots are exactly the eigenvalues of A.

That the roots are the eigenvalues is immediate: $\chi_A(\lambda) = 0$ iff $\lambda I - A$ is singular iff λ is an eigenvalue. The characteristic polynomial is moreover a property of the *operator*, not of the basis: for similar matrices,

$$\det(\lambda I - P^{-1}AP) = \det\bigl(P^{-1}(\lambda I - A)P\bigr) = \det(\lambda I - A),$$

so $\chi_{P^{-1}AP} = \chi_A$. Consequently eigenvalues are similarity invariants, and so are the two coefficients we already met in Lecture 4.

Proposition 5.17 (Trace and determinant from eigenvalues). Over $\mathbb{C}$, if $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A (listed with multiplicity, i.e. the roots of χ_A), then

$$\operatorname{tr} A = \lambda_1 + \cdots + \lambda_n, \qquad \det A = \lambda_1 \cdots \lambda_n.$$

Proof. Expanding $\chi_A(\lambda) = \det(\lambda I - A) = \lambda^n - (\operatorname{tr} A)\lambda^{n-1} + \cdots + (-1)^n \det A$ and comparing with $\prod_k(\lambda - \lambda_k) = \lambda^n - (\sum_k \lambda_k)\lambda^{n-1} + \cdots + (-1)^n \prod_k \lambda_k$ matches the coefficients. ■

Example 5.18 (A 2×2 characteristic polynomial). For $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ of Example 5.4, $\chi_A(\lambda) = (\lambda-2)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda-1)(\lambda-3)$, recovering the eigenvalues $1, 3$. Note $\operatorname{tr} A = 4 = 1+3$ and $\det A = 3 = 1 \cdot 3$. ◁

Example 5.19 (The 2×2 characteristic polynomial in general). For any $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$,

$$\chi_A(\lambda) = \det\begin{pmatrix} \lambda - a & -b \\ -c & \lambda - d \end{pmatrix} = \lambda^2 - (a+d)\lambda + (ad - bc) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A,$$

so the eigenvalues are $\frac{1}{2}\bigl(\operatorname{tr} A \pm \sqrt{(\operatorname{tr} A)^2 - 4\det A}\bigr)$. They are real and distinct, real and repeated, or a complex-conjugate pair according to the sign of the discriminant $(\operatorname{tr} A)^2 - 4\det A$—the rotation of Example 5.22 has negative discriminant whenever $\theta \notin \{0, \pi\}$. ◁

Example 5.20 (A 3×3 characteristic polynomial). For $A = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{smallmatrix}\right)$,

$$\chi_A(\lambda) = \det\begin{pmatrix} \lambda - 2 & 0 & 0 \\ -1 & \lambda - 3 & 0 \\ 0 & -1 & \lambda - 2 \end{pmatrix} = (\lambda-2)(\lambda-3)(\lambda-2) = (\lambda-2)^2(\lambda-3),$$

expanding along the top row of this lower-triangular matrix. The eigenvalues are 2 (with multiplicity 2) and 3. ◁

Example 5.21 (From eigenvalues to eigenspaces). For the same $A = \left(\begin{smallmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 0 & 1 & 2 \end{smallmatrix}\right)$ we find the eigenspaces. For $\lambda = 3$, $(A - 3I)v = 0$ forces $v_1 = 0$ and $v_2 = v_3$, so $E(3, A) = \operatorname{span}\big((0, 1, 1)\big)$. For $\lambda = 2$, $(A - 2I)v = 0$ gives $v_1 + v_2 = 0$ and $v_2 = 0$, hence $v_1 = v_2 = 0$ with v_3 free, so $E(2, A) = \operatorname{span}\big((0, 0, 1)\big)$ is only *one*-dimensional. The eigenvalue 2 thus has algebraic multiplicity 2 but geometric multiplicity 1: A is defective and cannot be diagonalized. ◁

Example 5.22 (Complex eigenvalues of a rotation). The rotation matrix $R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$ has $\chi(\lambda) = (\lambda - \cos\theta)^2 + \sin^2\theta = \lambda^2 - 2\cos\theta\,\lambda + 1$, with roots $\lambda = \cos\theta \pm i\sin\theta = e^{\pm i\theta}$. There are no real roots unless $\theta \in \{0, \pi\}$—confirming Figure 2—but over $\mathbb{C}$ the eigenvalues are the unit-modulus numbers $e^{\pm i\theta}$. ◁

A root of χ_A has an *algebraic multiplicity* (its multiplicity as a root) and a *geometric multiplicity* ($\dim E(\lambda, T)$). They can differ.

Proposition 5.23 (Geometric ≤ algebraic). For each eigenvalue, the geometric multiplicity does not exceed the algebraic multiplicity.

Proof. Let $g = \dim E(\lambda, T)$. Choose a basis $v_1, \dots, v_g$ of $E(\lambda, T)$ and extend it to a basis $v_1, \dots, v_n$ of V. Since $Tv_k = \lambda v_k$ for $k \le g$, the matrix of T in this basis is block upper-triangular,

$$\mathcal{M}(T) = \begin{pmatrix} \lambda I_g & B \\ 0 & C \end{pmatrix},$$

so by Definition 5.16 and the determinant of a block-triangular matrix,

$$\chi_T(t) = \det\big(tI - \mathcal{M}(T)\big) = (t - \lambda)^g \det\big(tI_{n-g} - C\big).$$

Hence $(t - \lambda)^g$ divides χ_T, so the algebraic multiplicity of λ is at least g, the geometric multiplicity. ■

When some eigenvalue has geometric multiplicity strictly less than its algebraic multiplicity, the operator has too few eigenvectors to form a basis; it is called *defective* and cannot be diagonalized. The shear $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ is the standard culprit: $\chi(\lambda) = (\lambda - 1)^2$ (algebraic multiplicity 2), yet $E(1, T) = \operatorname{span}\big((1, 0)\big)$ is only one-dimensional.

The fix is to relax what we ask of an eigenvector. *The next definition, theorem, and example—generalized eigenvectors—are a bridge to the Jordan material of Lecture 6 and are offered for self-study; the rest of the course does not depend on them.*

Definition 5.24 (Generalized eigenvectors and eigenspaces *[self-study]*). A nonzero v is a *generalized eigenvector* of T for λ if $(T - \lambda I)^k v = 0$ for some $k \ge 1$. The *generalized eigenspace* is

$$G(\lambda, T) = \{v : (T - \lambda I)^k v = 0 \text{ for some } k\} = \ker(T - \lambda I)^{\dim V},$$

and it contains the ordinary eigenspace $E(\lambda, T) = \ker(T - \lambda I)$.

Theorem 5.25 (Generalized eigenspace decomposition · cited). Over $\mathbb{C}$, $\dim G(\lambda, T)$ equals the algebraic multiplicity of λ, and V is the direct sum of the generalized eigenspaces,

$$V = G(\lambda_1, T) \oplus \cdots \oplus G(\lambda_r, T).$$

So a defective operator still has *enough* generalized eigenvectors to span the space, even when it lacks enough ordinary ones. This repairs the failure of diagonalization, and it founds the Jordan form of Lecture 6, which—like this decomposition—is stated there for use rather than proved, hence the cited tag.

Example 5.26 (A generalized eigenvector). For the shear $T = \left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ the only eigenvalue is 1, with the one-dimensional eigenspace $E(1,T) = \operatorname{span}((1,0))$. But $(T-I)^2 = 0$, so $G(1,T) = \ker(T-I)^2 = \mathbb{C}^2$ is the entire plane. The vector $(0,1)$ is a generalized eigenvector: $(T-I)(0,1) = (1,0)$ and $(T-I)^2(0,1) = 0$. The chain $(0,1) \to (1,0) \to 0$ is a *Jordan chain*, threading the eigenvector and the generalized eigenvector above it. ◁

Example 5.27 (A full diagonalization). The lower-triangular matrix $A = \left(\begin{smallmatrix}2&0&0\\0&3&0\\1&0&5\end{smallmatrix}\right)$ has the distinct eigenvalues $2,3,5$, so it is diagonalizable. Solving $(A-\lambda I)v = 0$ gives eigenvectors $(-3,0,1)$, $(0,1,0)$, $(0,0,1)$ for $\lambda = 2,3,5$ respectively. Collecting them as the columns of P,

$$P = \begin{pmatrix}-3&0&0\\0&1&0\\1&0&1\end{pmatrix}, \qquad P^{-1}AP = \begin{pmatrix}2&0&0\\0&3&0\\0&0&5\end{pmatrix} = \operatorname{diag}(2,3,5).$$

As a check, $\operatorname{tr} A = 10 = 2+3+5$ and $\det A = 30 = 2\cdot 3\cdot 5$, exactly as Proposition 5.17 requires.
◁

5 Upper-triangular matrices

Even when an operator cannot be diagonalized, over $\mathbb{C}$ it can always be brought to *upper-triangular* form, which is nearly as useful: the eigenvalues are displayed on the diagonal.

> **Proposition 5.28 (Triangular form encodes invariance).** Let $v_1,\dots,v_n$ be a basis of V. The matrix of T in this basis is upper triangular if and only if $Tv_k \in \operatorname{span}(v_1,\dots,v_k)$ for each k—equivalently, each $\operatorname{span}(v_1,\dots,v_k)$ is invariant under T.

Proof. The k-th column of the matrix expresses Tv_k in the basis; it has zeros below the diagonal exactly when Tv_k uses only $v_1,\dots,v_k$. The invariance reformulation follows because the spans are nested. ■

> **Theorem 5.29 (Triangularizability over $\mathbb{C}$ • sketched).** Every operator on a finite-dimensional complex vector space has an upper-triangular matrix with respect to some basis.

Proof sketch. Induct on $n = \dim V$. By Theorem 5.11 there is an eigenvalue λ and eigenvector v_1. Pass to the quotient $V/\operatorname{span}(v_1)$, of dimension $n-1$, on which T induces an operator; by induction it has a triangularizing basis $\bar v_2,\dots,\bar v_n$, and lifting these to $v_2,\dots,v_n \in V$ gives $Tv_k \in \operatorname{span}(v_1,\dots,v_k)$ for every k. By Proposition 5.28 the matrix in $v_1,\dots,v_n$ is upper triangular. The only place $\mathbb{C}$ is used is Theorem 5.11, to produce an eigenvalue at each step; over any field the same induction runs as soon as χ_T splits there, since a root of χ_T in the field is exactly an eigenvalue. ■

That last observation is used in Lecture 10, where the real spectral theorem needs triangularization over $\mathbb{R}$.

One consequence of triangular form is needed later, in the general proof of Cayley–Hamilton (Lecture 6).

Lemma 5.30 (Triangular products annihilate). Suppose T has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ with respect to a basis $v_1, \ldots, v_n$. Then

$$(T - \lambda_1 I)(T - \lambda_2 I) \cdots (T - \lambda_n I) = 0.$$

Proof. Upper-triangularity says $Tv_k \in \operatorname{span}(v_1, \ldots, v_k)$, so $(T - \lambda_k I)v_k \in \operatorname{span}(v_1, \ldots, v_{k-1})$. We show by induction on k that $(T - \lambda_1 I) \cdots (T - \lambda_k I)$ annihilates $v_1, \ldots, v_k$. For $k = 1$, $(T - \lambda_1 I)v_1 = 0$. Assume it for $k - 1$. All the factors are polynomials in T, hence commute, so we may apply $(T - \lambda_k I)$ to v_k first: it lands in $\operatorname{span}(v_1, \ldots, v_{k-1})$, which the remaining factors kill by hypothesis. The earlier basis vectors $v_1, \ldots, v_{k-1}$ are already annihilated by $(T - \lambda_1 I) \cdots (T - \lambda_{k-1} I)$. Taking $k = n$, the product annihilates a basis of V, so it is the zero operator. ■

The payoff is that for triangular matrices the spectrum is read off instantly.

Theorem 5.31 (Eigenvalues sit on the diagonal). If T has an upper-triangular matrix with diagonal entries $\lambda_1, \ldots, \lambda_n$ in some basis, then $\operatorname{spec}(T) = \{\lambda_1, \ldots, \lambda_n\}$. Moreover T is invertible iff no diagonal entry is 0.

Proof. With A upper triangular, so is $\lambda I - A$, and the determinant of a triangular matrix is the product of its diagonal entries:

$$\chi_A(\lambda) = \det(\lambda I - A) = (\lambda - \lambda_1) \cdots (\lambda - \lambda_n).$$

Its roots—the eigenvalues—are exactly $\lambda_1, \ldots, \lambda_n$. Taking $\lambda = 0$ gives $\det A = \lambda_1 \cdots \lambda_n$, which is nonzero iff no λ_k is 0, i.e. iff T is invertible. ■

Example 5.32 (Reading eigenvalues off a triangle). The matrix $\begin{pmatrix} 3 & 7 & -2 \\ 0 & 3 & 5 \\ 0 & 0 & -1 \end{pmatrix}$ is upper triangular, so without any computation its eigenvalues are $3, 3, -1$: that is, 3 with algebraic multiplicity 2 and -1 with multiplicity 1. Its determinant is $3 \cdot 3 \cdot (-1) = -9 \neq 0$, so it is invertible. ◁

Example 5.33 (Triangular form exposes a defective spectrum). The shear $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is already upper triangular, so by Theorem 5.31 its only eigenvalue is 1, repeated twice. It is not diagonalizable, as its eigenspace is one-dimensional, yet the triangular form still hands us the spectrum at a glance. This is why Theorem 5.29 remains valuable even when diagonalization is impossible. ◁

Example 5.34 (A two-level system: the Pauli matrix σ_x). The spin observable $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (in units of $\hbar/2$) has $\chi(\lambda) = \lambda^2 - 1$, so its eigenvalues are ± 1—the two possible outcomes of measuring spin along x. The eigenvectors are $\frac{1}{\sqrt{2}}(1, 1)$ for $+1$ and $\frac{1}{\sqrt{2}}(1, -1)$ for -1, the "spin-right" and "spin-left" states. Measuring σ_x returns one of its eigenvalues and leaves the system in the matching eigenvector—the physical meaning of the spectrum. ◁

Example 5.35 (Compatible and incompatible observables). The observable $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ has eigenvectors $(1, 0), (0, 1)$ for eigenvalues ± 1. These are *not* eigenvectors of σ_x, and correspondingly $\sigma_x \sigma_z \neq \sigma_z \sigma_x$. Commuting operators always share an eigenvector and can be simultaneously diagonalized; non-commuting ones such as σ_x and σ_z cannot, which is the linear-algebra origin of the uncertainty principle for incompatible observables. ◁

Remark 5.36 (Observables and energy levels). In quantum mechanics the eigenvalue equation $\hat{H}\psi = E\psi$ is the time-independent Schrödinger equation: the eigenvalues E are the allowed energies and the eigenvectors ψ are the stationary states. Because physical observables are represented by self-adjoint operators (Lecture 9), their eigenvalues turn out to be real and their eigenvectors can be chosen to form an orthonormal basis—the spectral theorem of Lecture 10. The abstract existence theorem above is what makes those measured values exist in the first place.

Remark 5.37. The discreteness of atomic spectral lines is, mathematically, the discreteness of a Hamiltonian's eigenvalues: bound-state energies form a separated set because the relevant operator has a discrete spectrum—a phenomenon made precise for infinite-dimensional spaces in Lectures 13–14.

Summary of main concepts

An eigenvector is a direction an operator only rescales, and the scale factor is its eigenvalue; the eigenspace $E(\lambda, T) = \mathrm{null}(T - \lambda I)$ collects them, and λ is an eigenvalue exactly when $T - \lambda I$ fails to be invertible. Eigenvectors for distinct eigenvalues are independent, so an n-dimensional space admits at most n eigenvalues, and n distinct ones force a diagonalizing basis. Over $\mathbb{C}$ every operator has an eigenvalue—the fundamental theorem of algebra in disguise—while over $\mathbb{R}$ it may have none. The characteristic polynomial $\chi_A(\lambda) = \det(\lambda I - A)$ has the eigenvalues as its roots, is a similarity invariant, and yields $\operatorname{tr} A = \sum \lambda_k$, $\det A = \prod \lambda_k$. Finally, over $\mathbb{C}$ every operator is upper-triangularizable, and then its eigenvalues are simply the diagonal entries.

- **Eigenvalue, eigenvector, eigenspace**—$Tv = \lambda v$; $E(\lambda, T) = \mathrm{null}(T - \lambda I)$; λ is an eigenvalue $\iff T - \lambda I$ is not invertible.
- **Independence of eigenvectors**—distinct eigenvalues give independent eigenvectors; at most $\dim V$ of them.
- **Existence over** $\mathbb{C}$—every operator on a complex space has an eigenvalue (fundamental theorem of algebra).
- **Characteristic polynomial**—$\chi_A(\lambda) = \det(\lambda I - A)$; its roots are the eigenvalues, with $\operatorname{tr} A = \sum \lambda_k$, $\det A = \prod \lambda_k$.
- **Upper-triangular form**—exists over $\mathbb{C}$; the eigenvalues are then the diagonal entries.
- **Generalized eigenvectors**—the generalized-eigenspace decomposition behind Jordan form (self-study).

Exercises for this lecture are in the companion sheet `exercises-5.pdf`.